

Chemical exposure early in life and the neurodevelopment of children--an overview of current epidemiological evidence.

.

A number of chemicals have been shown to demonstrate neurotoxic effects either in human or laboratory animal studies. This article aims at evaluating the impact of exposure to several chemicals including: organophosphate, organochlorine pesticides, polychlorinated biphenyls (PCBs), mercury and lead on the neurodevelopment of children by reviewing the most recent published literature, and answer the question whether any progress has been made in the epidemiology of the neurodevelopment of children induced by exposure to those chemicals. The result of the presented studies show that exposure to the above-mentioned chemicals may impair the neurodevelopment of children. Neonates exposed to organophosphate pesticides demonstrated a higher proportion of abnormal reflexes, and young children had more attention problems. Exposure to organochlorine pesticides in children was associated with alertness, quality of alert responsiveness, cost of attention and other potential attention associated measures. The majority of studies indicate the negative impact of lead exposure at the level $<10\text{ }\mu\text{g/dl}$ or even $<5\text{ }\mu\text{g/dl}$ on the neurodevelopment of children. The results of studies on exposure to PCBs, mercury, and their effect on neurodevelopment are inconsistent. Some suggest that prenatal exposure to PCBs and mercury is related to performance impairments, attention and concentration problems, while other do not present any statistically significant association. The studies were mostly well designed, using prospective cohorts with the exposure assessment based on the biomarker of exposure. Concerning the covariates and confounders affecting the endpoints in most of the presented studies, confounders were included in data analysis. In order to recognize the early cognitive, motor and language outcomes of chemical exposures, well standardized tools were used for evaluating the neurodevelopmental effects and offer an early and fairly comprehensive measure of child development. Because the neurotoxicants may cross the placenta and the fetal brain, exposure consideration regarding the reduction of exposure to those chemicals should be implemented.